

The Industrialization of Sub-Saharan Africa

Zhihu column · 2026-05-17

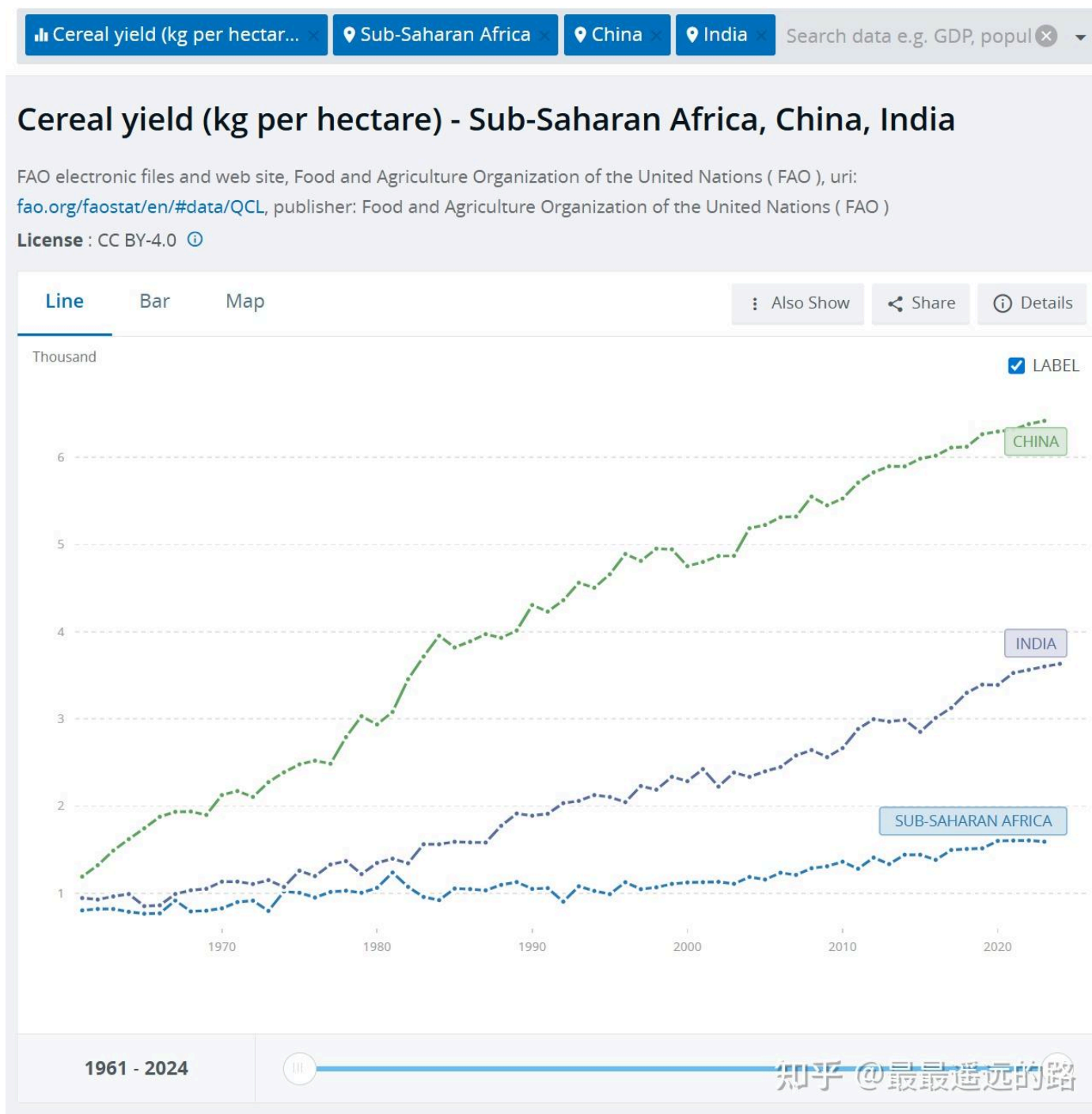
In my series of articles on the capitalist world-system ("The Rise and Fall of the Capitalist World-System (III): An Analysis of the Contradictions of the Three Stages of the World-System"), I proposed: "The center of global working-class struggle in the 2010s was in East Asia; in the 2020s it moved to Southeast Asia; in the 2030s it will move to South Asia and North Africa; in the 2040s it may move to East Africa; in the 2050s, to West Africa. When large-scale working-class struggle begins in West Africa, industry will have nowhere left to go; the struggle will spread back from the periphery to the center, and the wave of class struggle will make a historic return on a global scale." The shifting of the center of working-class struggle means the shifting of the center of manufacturing and the shifting of the exploitation center of the global capitalist system. But many people are skeptical about whether Africa can industrialize and whether African capitalism can develop. They believe that because of race, cultural traditions, AIDS, imperialist economic colonization, China's "welding shut" the door of industrialization, and various other reasons, Africa cannot complete industrialization. And even those who think African capitalism can develop believe Africa's industrialization lies a hundred years in the future — not something we, the living, need to concern ourselves with. Is that really so? Is the capitalist industrialization of Africa a distant dream in the present world-system? Let us look together at how far Africa's industrialization has progressed, and whether the prediction that African industrialization takes off by 2040 is credible.

Since North Africa started industrializing early and is now in the stage of rapid industrialization, likely completing it within about 20 years, this article focuses mainly on sub-Saharan Africa.

I. Agricultural Modernization in Sub-Saharan Africa

(1) The State of the Agricultural Sector

The figure below ^[1] shows the historical trend of cereal yields in China, India, and sub-Saharan Africa. In 1961, yields per hectare in China, India, and sub-Saharan Africa were roughly the same. From 1960 to 1980, China's agricultural yields rose nearly fourfold. India's yields rose nearly twofold from 1990 to 2010. Africa has also grown, but slowly.



Behind the great increase in agricultural productivity lie fertilizer and agricultural machinery. The figure below ^[2] shows fertilizer use per hectare of arable land in China, India, and sub-Saharan Africa. In 2022, sub-Saharan Africa used 18.5 kg of fertilizer per hectare — equivalent to China in the 1960s and India in the 1970s.

Fertilizer consumption (kilograms per hectare of arable land) - Sub-Saharan Africa, China, India

FAO electronic files and web site, Food and Agriculture Organization of the United Nations (FAO), publisher: Food and Agriculture Organization of the United Nations (FAO)

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The figure below ^[3] shows agricultural employment as a share of total employment in China, India, and sub-Saharan Africa. In 1990, the agricultural employment shares of China, India, and black Africa were similar; but the rapid rise in China's agricultural productivity before the 1980s left the countryside with a huge surplus labor force, ensuring an ample labor supply during the twenty years of rapid capital accumulation from 1990 to 2010.

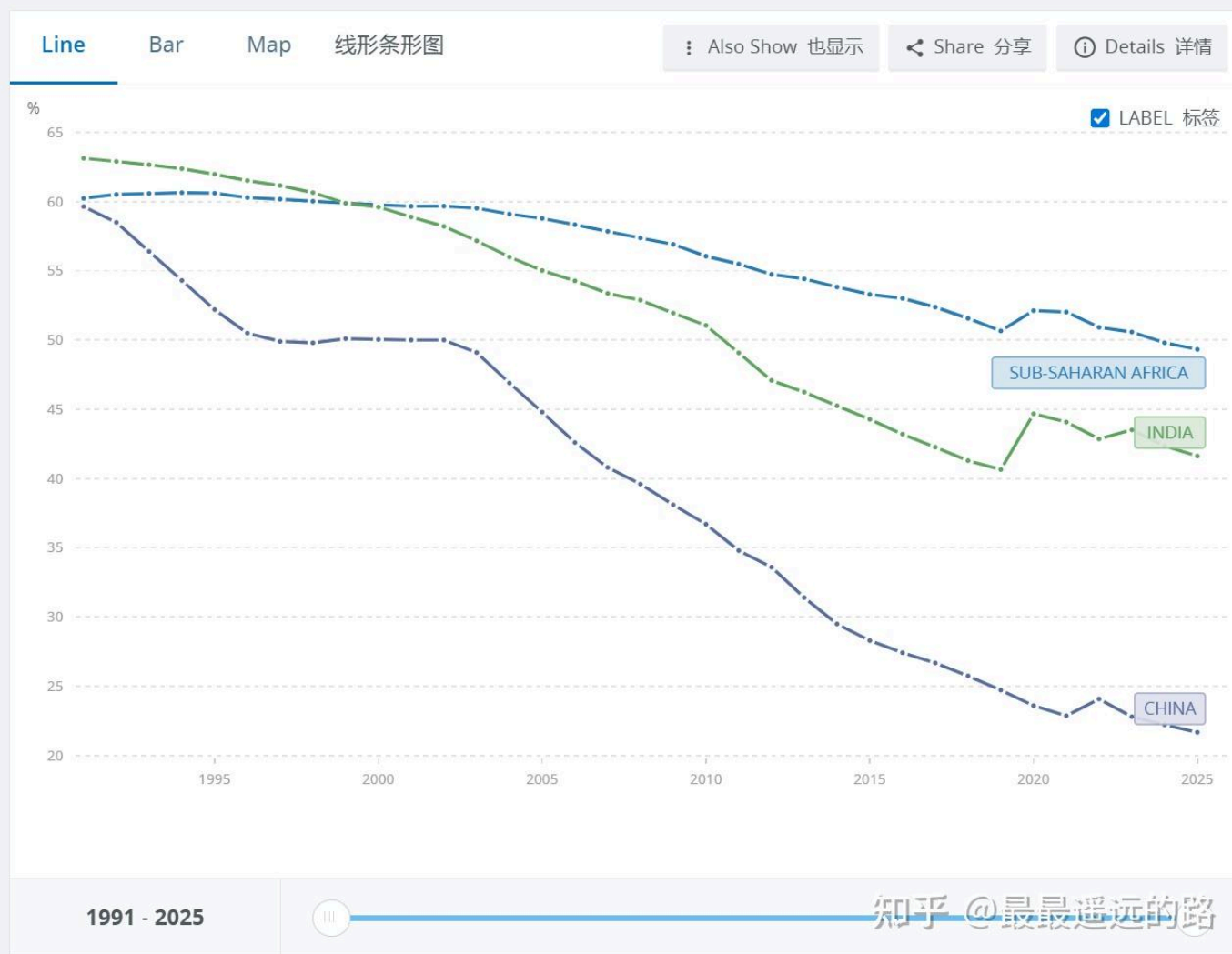
Employment in agriculture (% of total employment) (modeled ILO estimate) - Sub-Saharan Africa, India, China

农业就业（占总就业百分比）（国际劳工组织模型估计）-撒哈拉以南非洲、印度、中国

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

国际劳工组织模型估计数据库 (ILOEST) , 国际劳工组织 (ILO) , uri: ilostat.ilo.org/data/bulk, 出版者: ILOSTAT, 类型: 外部数据库, 访问日期: 2026 年 1 月 17 日

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(2) Large-Scale Modernization of African Agriculture Is About to Begin

The history of almost all industrialized countries shows that agricultural modernization is an important precondition of industrialization. Without developed agriculture, the light industries that process farm products lack raw materials. Food shortages force large amounts of foreign exchange to be spent on grain imports, slowing industrialization. Moreover, unless agricultural productivity rises

rapidly, the countryside cannot provide the surplus population for transfer to the industrial sector. According to the earlier analysis, African agricultural modernization has not yet begun, so the takeoff of industrialization is even further out of the question. But the thesis that industrialization must wait for agricultural modernization is a summary of the experience of the era of national industrial capitalism. In the transnational capitalist stage, although agricultural modernization remains very important as a support for industry, this strict sequence can no longer be observed. In fact, modern industry and modern agriculture can develop simultaneously and reinforce each other. Industry we will discuss in the next chapter; here we continue with agricultural modernization. In my view, the modernization revolution of African agriculture is about to begin — or rather, has actually already begun.

1. Fertilizer

Before the 2010s, sub-Saharan Africa had essentially no fertilizer capacity. Taking urea as an example: before the 2010s, only South Africa and Nigeria had small urea capacity, utterly unable to meet local farmers' needs. Indorama Group (an Indian firm) took over a fertilizer plant from the government and installed a new line; after starting production in 2016 its urea capacity reached 1.4 million tonnes, still unable to meet the country's demand of over 3 million tonnes. In 2021, Africa's richest man, Dangote, brought a new 3-million-tonne urea plant into production. Notably, China's urea output in 1978 was only 3.788 million tonnes ^[4]. So this single Dangote plant's urea capacity roughly matches China's entire annual level of the late 1970s.

At present, the Dangote urea plant is being expanded, in cooperation with a German company ^[5], raising the capacity of this factory in the Lekki free trade zone to 9 million tonnes. At the same time, a new 3-million-tonne urea plant is to be built in Ethiopia. When these two projects come online, the Dangote group's total urea capacity will reach 12 million tonnes, making it one of the world's largest fertilizer producers ^[6]. Indorama also plans to expand its Nigerian urea capacity to 4.2 million tonnes ^[7]. If the projects are completed, the two companies' total Nigerian urea capacity will reach about 13 million tonnes. China's fertilizer output in 2000 was 53.687 million tonnes ^[4]; adjusted by population, Nigeria has already reached or surpassed China's 2007 level. Besides Nigeria, in recent years several African countries have successively brought fertilizer plants online or planned projects ^{[8] [9] [10] [11]}. Most of these projects are in the hundreds of thousands to a million tonnes, but the sight of fertilizer plants sprouting everywhere shows that African industrial development has arrived at this stage. China's industrialization has lowered the cost of building fertilizer plants in Africa, speeding up the commissioning of new capacity. China's urea output in 1995 was about 20 million tonnes; sub-Saharan Africa's population is about the same as China's, so reaching China's 1995 level within ten years should not be a problem. Between 2040 and 2045, it should reach China's level around 2000.

2. Agricultural Machinery

Africa's current agricultural mechanization rate is very low. Taking tractors as an example ^[12], the fleet is far below the international average.

指标	非洲水平	全球平均水平	亚洲水平 (不含中国)	数据来源
拖拉机保有量	13台/万公顷 (撒哈拉以南)	200台/万公顷	89台/万公顷	联合国粮农组织 (FAO) 2024
灌溉设施覆盖率	7%	31%	28%	非洲农业发展银行 (AFDB) 2024
谷物收割机械化率	不足10%	68%	45%	联合国粮农组织 (FAO) 2024
农机年缺口金额	50-60亿美元	-	-	Egatee非洲商机分 析2025
2030年市场规模预测	2030年将突破 200亿美元	-	-	Egatee非洲商机分 析2025
农业人口占比	70% (依赖农业收入)	40%	55%	Invest Africa 2024

表1：非洲农业机械化核心数据与全球对比

Source: "A trillion-yuan blue ocean awaiting cultivation: how Chinese agricultural machinery is rewriting Africa's agricultural map through cross-border e-commerce"

At present, several African countries have started building tractor assembly plants — some in production, some under construction or planned.

The YTO Congo (Brazzaville) agricultural equipment assembly plant project is a key Belt and Road project followed personally by the president of Congo (Brazzaville), built jointly by YTO and a local enterprise, mainly producing 70–140 horsepower tractors. When complete, it will become one of the most advanced tractor production lines in African countries, with an annual capacity of 3,000 tractors, well meeting the agricultural machinery needs of the local area and neighboring countries. The project also creates many local jobs and will greatly enhance the country's machinery manufacturing capacity and agricultural mechanization level. ^[13]

Iran will build a tractor assembly plant in Tanzania. Recently, witnessed by Tanzanian Foreign and East African Community Minister Mahmoud Thabit Kombo and Iranian Agriculture Minister Gholamreza Nouri, the two countries' public and private sectors signed more than a dozen cooperation MOUs in Dar es Salaam; the two officials jointly committed to strengthening bilateral cooperation. Tanzanian Deputy Agriculture Minister David Silinde said Iran's decision to build a tractor plant in Tanzania fits President Samia Suluhu Hassan's plan to provide over 10,000 tractors to farmers by 2030. Tanzania will allocate 10,000 acres of land to Iranian investors. ^[14]

On September 5, representatives of the Ethiopian Engineering Group and YTO Sino-Africa Machinery laid the foundation for an assembly plant in Mojo with an annual capacity of 10,000 tractors. ^[15]

German automaker Volkswagen announced plans to build an electric tractor manufacturing plant in Nigeria, an important step for Nigerian agricultural mechanization. ^[16]

According to the Angolan newspaper Jornal de Angola, the tractor assembly plant in the Luanda-Bengo Special Economic Zone (ZEE), with total investment of about \$100 million from the UAE, has rolled out its first tractors and agricultural implements suited to the Angolan market. As a representative industry of "Made in Angola," the plant has an annual capacity of about 3,000 tractors, with UAE financing and technical support from the well-known agricultural machinery company Massey Ferguson. ^[17]

Of course, local manufacturing is still very rudimentary; Africa's need for mechanization must be met by direct imports from the industrial countries. Chinese enterprises are very active here. Chinese firms facing a saturated domestic market are actively opening the African market:

Walking tractors, micro-tillers, mountain weeders, monorail transporters, tracked garden carriers... the reporter noted on site that these representative Zhejiang agricultural machinery firms, in selecting products for the African market, all chose the "classic models" in their product lines — more technically mature, more cost-controllable, applicable to more scenarios. "With their highly competitive products, Chinese agricultural machinery manufacturers are steadily strengthening their influence in the African market," affirmed an official of an African country's agriculture department. Statistics show Zhejiang is the main force: agricultural machinery exports to Africa grew from 139 million yuan in 2018 to 332 million yuan in 2023, an average annual increase of nearly 20 percent. In the first three quarters of 2024, Zhejiang's agricultural machinery trade with Egypt, South Africa, Nigeria, Algeria, Tanzania, Kenya, and Côte d'Ivoire each exceeded 15 million yuan, driving Zhejiang's total agricultural machinery exports to Africa up 45.4 percent year on year, to 348 million yuan. ^[18]

In 2017, Nongfu Machinery's self-developed NF902 tractor was first sold to Ghana. Export manager Huang Yan remembers it vividly: "When our machine roared across the Ghanaian fields, the crowd watching burst into cheers." This "iron ox of the fields," designed for the local "mostly plateau, locally undulating" terrain, raised work efficiency by 20 percent and was widely reported by local media. In the first quarter of this year, Hunan Nongfu Machinery's exports to Africa grew 20 percent year on year. "Our customized machinery is running Chinese speed across Africa," said General Manager Liao Zengcai proudly. This company, the first in China to develop lightweight tracked tractors, has sold its products to more than 20 countries, including Nigeria, Zambia, and Egypt. ^[19]

It is not only China: Indian firms are also actively developing agricultural machinery adapted to the African market, with very good cost-performance ^[20]. In recent years, African governments have vigorously promoted agricultural mechanization — the Ghanaian government imports tractors from Belarus and other countries ^[21]; the Ethiopian government plans to more than triple its agricultural machinery fleet ^[22]. The efforts of African countries to raise mechanization have had results: for example, the share of mechanized cultivation in Ethiopia has risen to 27 percent (the statistical basis is unclear). But considering that China's overall mechanized cultivation rate had only just exceeded 30 percent in 2000, African countries have already made no small achievement. Over the next fifteen years, through a combination of imports and local manufacturing, raising the mechanized share to about 40 percent (India's current level) should be no problem.

3. Seeds and Pesticides

Sub-Saharan African countries have also carried out their own breeding research, with some results. For example, the Ethiopian Institute of Agricultural Research (EIAR) developed the hybrid maize variety BH 6241, yielding 6–10 tonnes per hectare in trials — not advanced, but a usable seed suited to local planting ^[23]. Other countries have similar results with sorghum and maize hybrids.

Chinese hybrid rice varieties were also promoted in Africa long ago. And African countries promote varieties among themselves.

Many new rice varieties developed by Chinese scientists have taken root in many countries. For example, the International Green Super Rice project led by Chinese agricultural scientists has been trialed and promoted in 18 African and Asian countries, benefiting more than 1.6 million farm households. In Volhis's view, African food self-sufficiency is not an unattainable goal. He told reporters that Ethiopia used to spend \$800 million to \$1 billion a year importing wheat — now that is history; within just five years it achieved wheat self-sufficiency. He further analyzed that the key to Ethiopia's achievement in such a short time was promoting advanced agricultural technology to farmers, providing disease-resistant seeds, and building effective disease monitoring systems. ^[24]

Zimbabwe and Zambia have successfully built strong hybrid maize systems that West Africa can adopt. The International Institute of Tropical Agriculture (IITA) and CIMMYT recently launched a new Gates Foundation-funded program called "African Resilient Maize Hybrids" (2026–2028), aimed at helping farmers shift from low-yield open-pollinated varieties to high-yield, climate-resilient maize hybrids. ^[25]

On pesticides, in the past two years Chinese firms have opened several factories in Africa, starting localized production. Other countries may have similar firms.

On May 11, 2026, at Nigeria's Lekki Free Trade Zone, Essence Crop Science Nigeria FZE, under the Aijin Group, held its grand opening ceremony. This marks the official operation of the group's first

overseas pesticide manufacturing base, and means the group's globalization strategy has taken a substantial step from "product export" toward "local production, local technology, local service." [26]

Since Xin'an Co. established its Xin'an Sunshine subsidiary in Ghana, adhering to the philosophy of "local, professional, and authentic," it has invested over \$100 million in total and built Africa's first modern pesticide formulation plant — now the largest pesticide supplier in West Africa. Local employees account for over 90 percent of the workforce; more than 100 agricultural training sessions are held every year; and the high-yield corn solution has increased local per-mu yields by over 30 percent, earning Ghana's "Outstanding Contribution to Agricultural Development" award. [27]

At dawn, in Kumasi, Ghana, a group of Chinese builders carry their tools toward the construction site. They are the vanguard of Xin'an's push into Africa — a team that has gone to Ghana and Nigeria eight times, writing a legend on foreign soil with shovels and welding torches, building a modern base with an annual output of 30,000 tonnes of glyphosate from the ruins, achieving Nigeria's zero breakthrough in this field. [28]

II. Infrastructure in Sub-Saharan Africa

(1) Modernizing Transport Infrastructure

"To get rich, build roads first." This is a famous Chinese saying. Africans, for the sake of industrialization, are also developing roads and railways as a priority.

Below we mainly discuss East Africa and West Africa — the two regions that will be Africa's main industrial zones.

1. Railway construction in East Africa

East Africa is focusing on building the SGR network, linking the landlocked countries of Uganda, Rwanda, South Sudan, and Burundi to the ports of Kenya and Tanzania.

- The Northern Corridor runs from Kenya's port of Mombasa via Nairobi, Naivasha, Kisumu, and Malaba (Kampala, Uganda's capital) then connects to Rwanda, Burundi, DR Congo, and South Sudan. The Northern Corridor generally runs passenger trains at 120 km/h and freight at 80 km/h.
- The Central Corridor runs from Tanzania's port of Dar es Salaam to Tabora, Isaka, and Kigoma, then connects to Rwanda, Burundi, and DR Congo. The Central Corridor generally runs passenger trains at 160 km/h and freight at 120 km/h.

Kenya has already completed the Mombasa–Nairobi and Nairobi–Naivasha sections. The Naivasha–Kisumu–Malaba section is under construction by a Chinese company, and Uganda's Kampala–Malaba section is under construction by a Turkish company. Both projects are planned for completion within 2028. When open, Uganda's capital will have a direct line to Mombasa, cutting travel from over a day to 8–10 hours.



Source: https://www.researchgate.net/figure/Proposed-Kenyan-SGR-Network-Note-The-route-marked-grey-indicates-incomplete-phases-of_fig1_362568692

By comparison, Tanzania is moving faster — its line is nearly finished. Rwanda is therefore considering starting with the line connecting to Tanzania, since the Uganda–Kenya side is too slow. Tanzania's railway was built by a Chinese company, but the trains were bought from South Korea [29]. When the whole line opens, travel from Rwanda's capital to Dar es Salaam port will fall from 38 hours to 10–15 hours. Whether the 120 km/h of the northern line or the 160 km/h of the southern line, the starting point is already high. By comparison, in China in 2000, after the third speed-up, the national average passenger train speed was 60 km/h (I remember riding trains in 2006 — under 1,000 km took 15 hours), key lines ran at 120 km/h, and only a very few trunk lines (like the Beijing–Shanghai line) reached 160 km/h. Since the populations of these East African countries are mainly concentrated within a hundred-plus kilometers of the railway lines, once the East African SGR

network is completed in the 2030s, the rail travel experience in East Africa will directly approach China's 2000s — a leap in one step.



Source: <https://www.ntu.edu.sg/cas/news-events/news/detail/korean-made-high-speed-trains-make-debut-in-tanzania-as-work-on-the-usd10bn-sgr-project-picks-up-speed>

Besides the two SGR corridors, another major project is LAPSSET (Lamu Port and Lamu–Southern Sudan–Ethiopia Transport Corridor). From the new Lamu port in Kenya, through Isiolo, one line goes to South Sudan and another to Ethiopia. The corridor includes rail, road, and oil-and-gas pipelines. Since Kenya's population is concentrated in the south, this corridor mainly provides landlocked neighbors with access to the sea, and for Kenya itself, serves to develop the backward northern regions. This project currently has serious financing problems; the priority is building the port and roads, and the full railway will not be completed until the late 2030s or early 2040s.

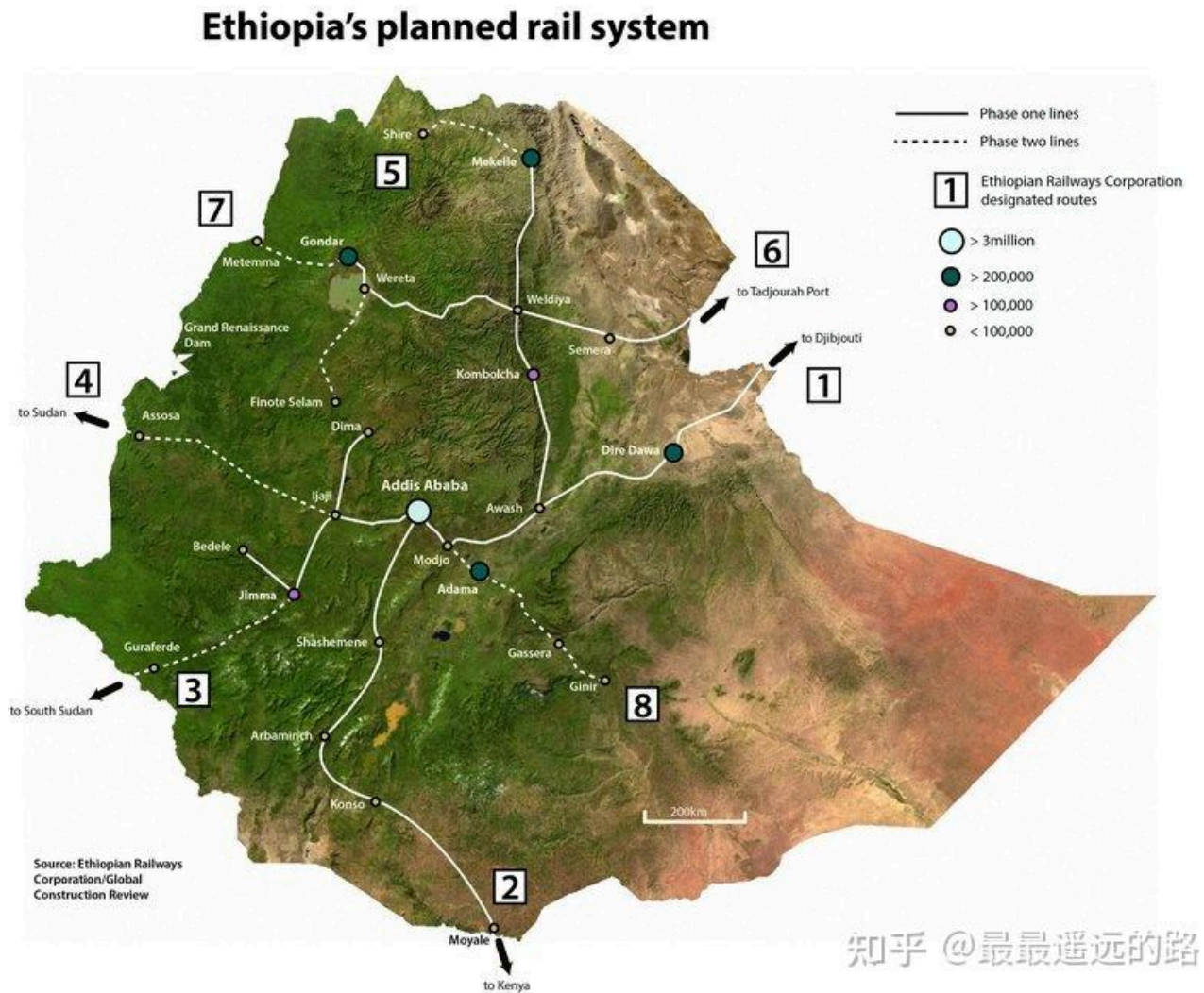


Map of the LAPSSET Corridor. LAPSSET ¼ Lamu Port-South Suda-Ethiopia Transport. Source: <http://www.lapsset.go.ke>.

Source: https://www.researchgate.net/figure/Map-of-the-LAPSSET-Corridor-LAPSSET-14-Lamu-Port-South-Suda-Ethiopia-Transport-Source_fig1_361579009

Ethiopia's railways mainly develop in four directions. The first is the line from the capital to Djibouti — already completed and in use (passenger 120 km/h, freight 80 km/h), solving the problem of access to the sea. The second connects cities around the capital. The third connects the major cities of

the northern population concentration (under construction, but interrupted by war; the project is being restarted). The fourth runs south to connect Kenya, as the first part of LAPSSSET, eventually reaching Lamu port. Completion of this line is estimated to be quite far off.



Source: <https://ethiopianembassy.be/?p=7681>

2. Road construction in East Africa

Expressway projects:

- The Mombasa–Nairobi expressway, about 500 km, four lanes both ways, will cut travel between the two cities from the current 6–10 hours to about 4 hours. Construction is expected to start within this year.
- The Kikira–Kisumu expressway, also called the Kenya–Uganda highway, 256 km, four lanes both ways, will cut travel time between the two cities from 6–10 hours to about 4 hours. Work may start at the end of 2026.

- The Nairobi–Nakuru–Mau Summit road, 243 km, an existing route being upgraded to four lanes both ways. Expected completion mid-2027.

There is also an expressway the Kenyan prime minister just proposed, from Mau Summit via Kisumu to Malaba ^[30]. It is said construction may begin within a year or two.

Basically, Kenya's expressways align with the SGR Northern Corridor: from Mombasa via Nairobi to Kisumu, then on to Uganda. When these lines are finished, the medium and large cities of Kenya's populated areas will all be linked by expressway.

Kenya also has two important non-expressway routes:

- The Lake Victoria ring road, passing the lakeside metropolis of Kisumu, nearly 500 km, two lanes both ways.
- The Isiolo–Mandera road, about 748 km, two lanes both ways, supported by the World Bank.



The Isiolo-Mandera Regional Road Corridor. © Africa Intelligence/Indigo Publications 2020

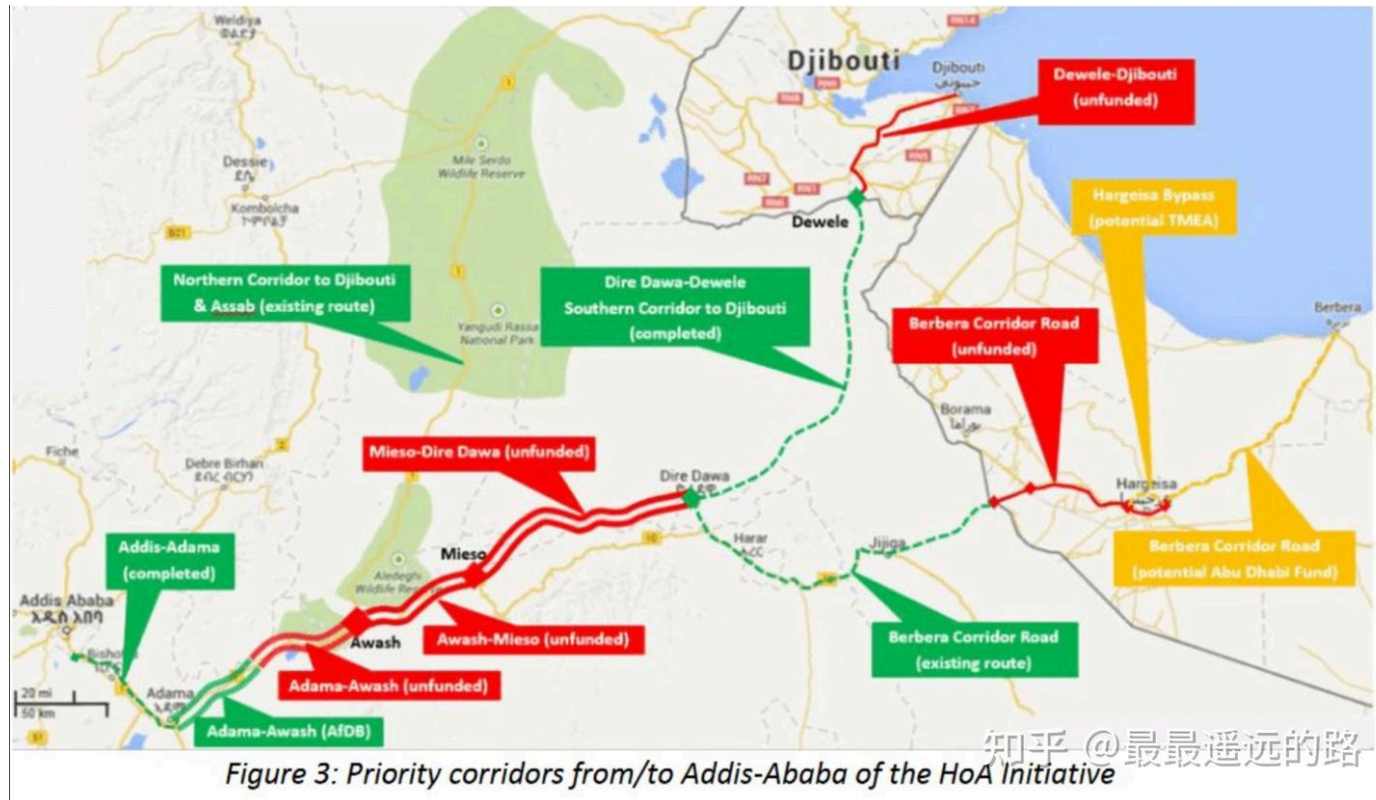
知乎 @最最遥远的路

Source: <https://www.africaintelligence.com/eastern-africa-and-the-horn/2020/12/14/nairobi-makes-headway-in-improving-access-to-northeast,109627229-art>

Tanzania's expressways are mostly short sections; the country's current priority is railways. Uganda is similar.

Ethiopia, however, is developing expressways, following a logic similar to its railways: priority to the capital–Djibouti route, then the northern routes. The figure below is an older schematic and its status is inaccurate. The Awash–Adama and Awash–Mieso–Dire Dawa sections, both four lanes, are already under construction. The Dire Dawa–Dewele section, originally two lanes, is being upgraded

to four. Essentially the whole capital–Djibouti corridor will be completed and opened within the next few years, four lanes throughout.

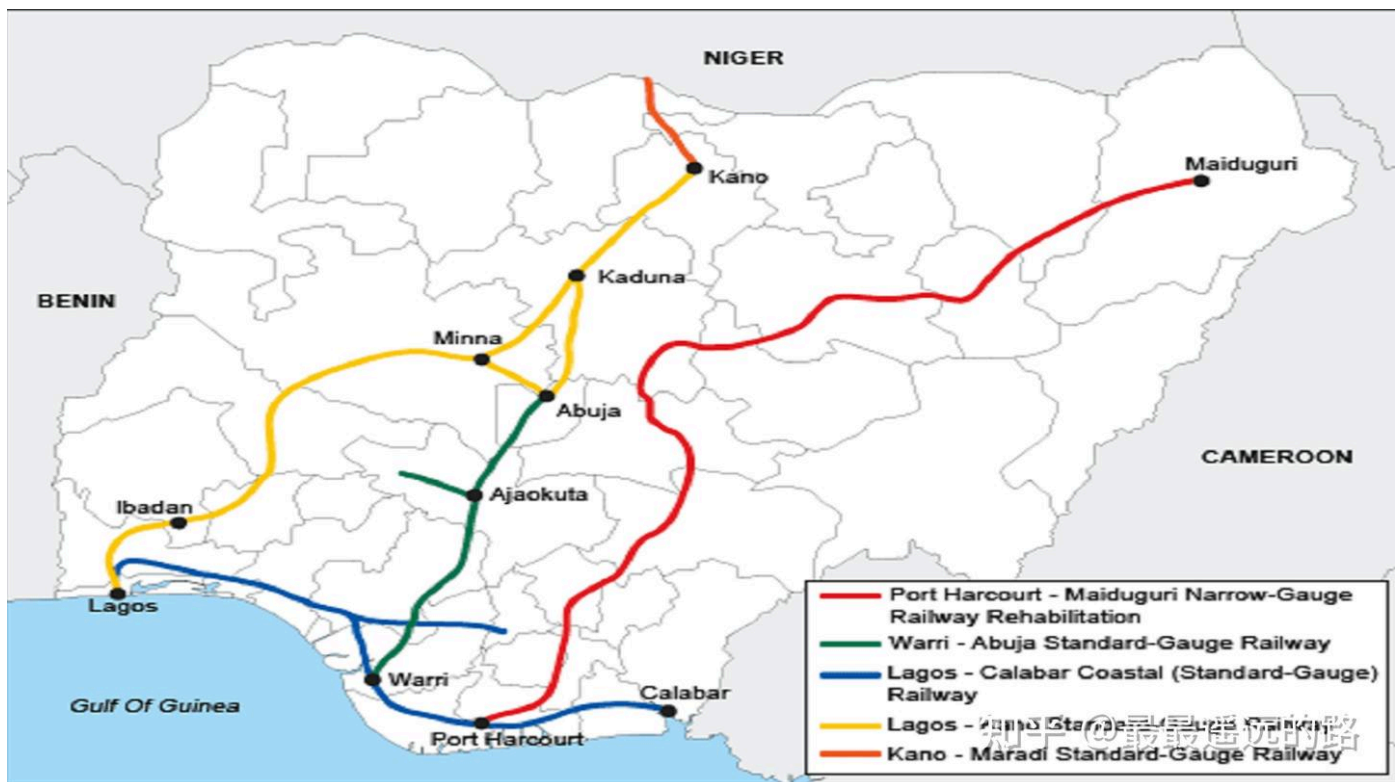


Source: <https://www.facebook.com/photo/?fbid=665860362245870&set=a.640726111425962>

On the whole, Ethiopia, Kenya, and Tanzania are investing heavily in transport; Tanzania focuses on rail, Ethiopia and Kenya on both. By coverage, Kenya is probably stronger, given its higher level of economic development.

3. Railway construction in West Africa

The key country of West Africa — Nigeria — has launched a grand railway plan, with five projects going simultaneously, totaling \$26 billion, almost all built by Chinese companies.



Source: <https://dmarketforces.com/nigerias-top-5-railway-projects-gulp-26-billion-report/>

- The line from the northern hub Kano to the largest city Lagos, standard gauge, over 1,100 km, at 150–200 km/h.
- The Itakpe–Abuja–Ajaokuta (site of a great steel plant awaiting revival)–Warri line, 326 km, standard gauge, 150 km/h.
- The Lagos–Calabar coastal railway, standard gauge, 1,400 km, 120–150 km/h.
- The Port Harcourt–Maiduguri narrow-gauge upgrade, 1,400 km, speeds raised to 80–100 km/h.

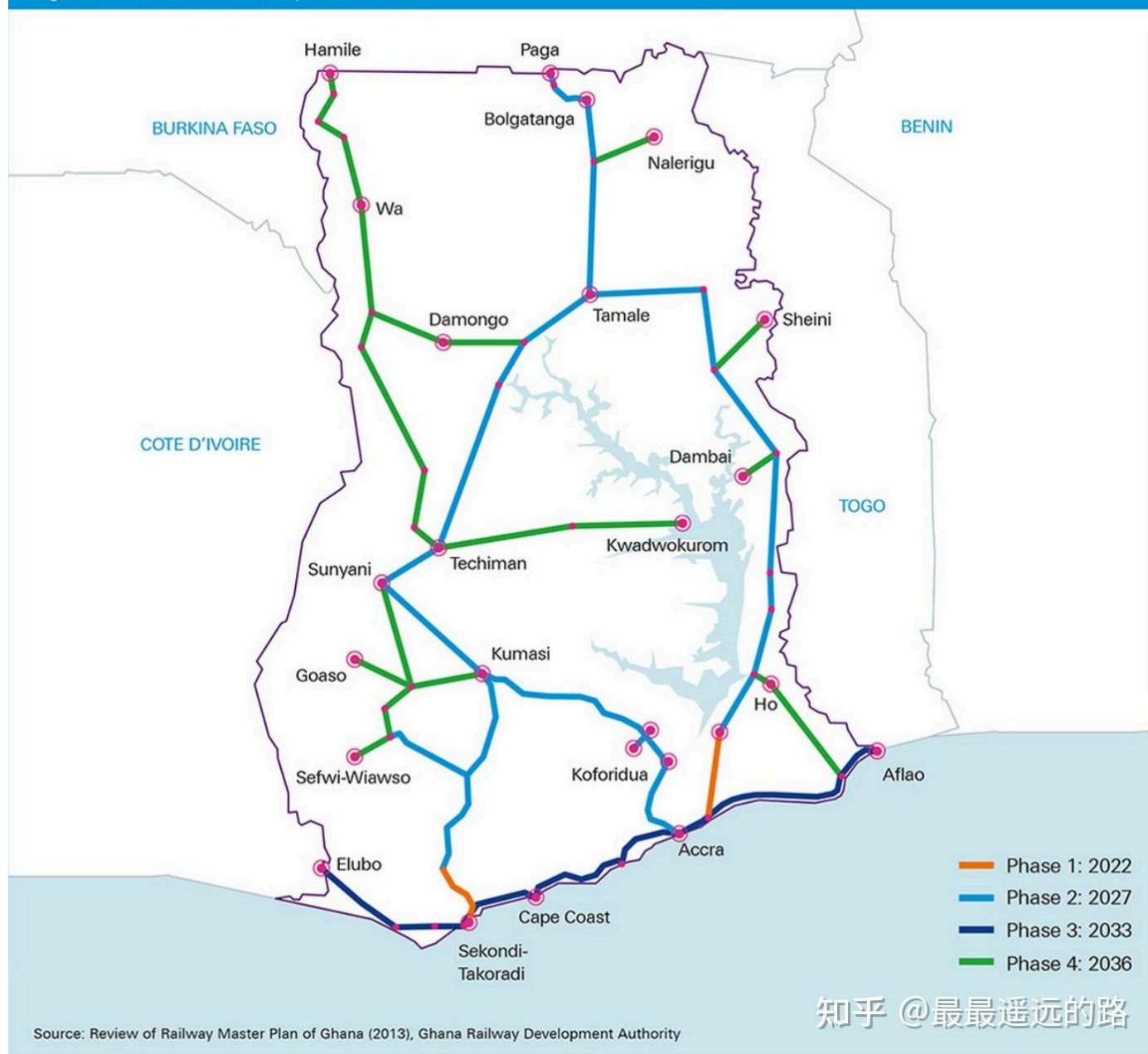
And one international line:

- From Kano to Maradi in Niger, standard gauge, 393 km, passenger 120 km/h (up to 160 km/h), heavy freight 80 km/h, normal freight 100 km/h.

All of these projects will be completed before 2030. When fully delivered, Nigeria's major cities will essentially be linked by rail lines running at 120–150 km/h, with travel experience and logistics speeds comprehensively upgraded.

The Ghanaian government has also formulated a grand railway development plan.

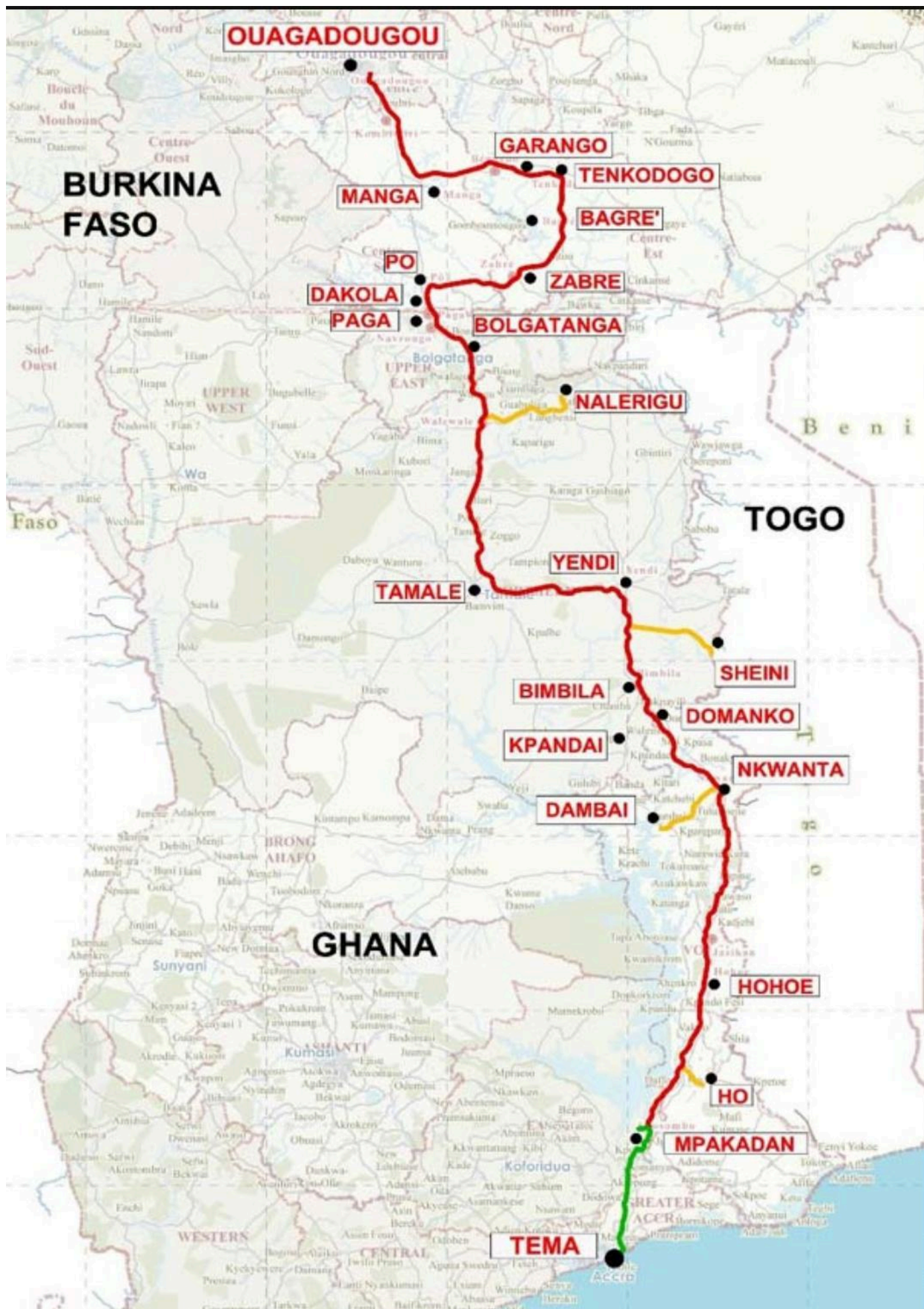
Figure 1: The Ghana Railway Master Plan

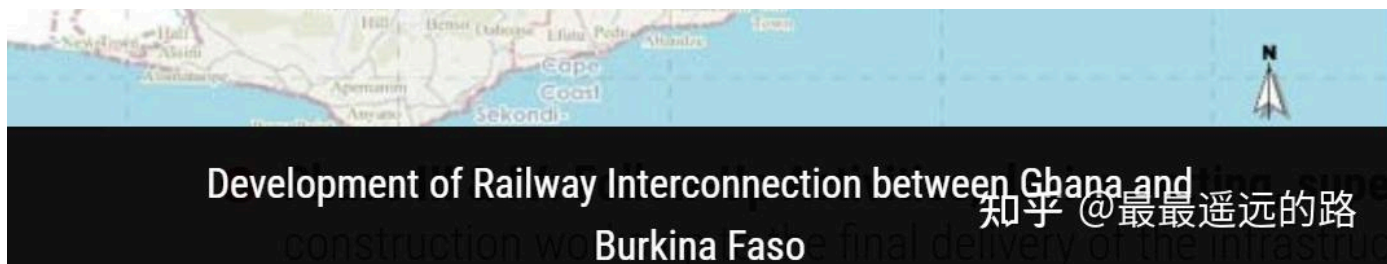


Source: <https://flow.db.com/topics/trade-finance/transforming-ghana-s-railway-infrastructure#!>

The line from the capital Accra to the second city Kumasi is fully designed: 300 km, design speeds of 120 km/h passenger and 100 km/h freight, an electrified double-track line.

The line from Tema via the northern third city Tamale to Ouagadougou in Burkina Faso is partly built (green); the rest is in feasibility study. The full line is 1,200 km at 120 km/h.





Source: <https://teamgroup.it/project/railway-interconnection-ghana-burkina-faso/>

The railway plans of the francophone West African countries center on a loop connecting five countries — a line of over 3,000 km, passenger 120 km/h, freight 80 km/h, including:

- Abidjan–Ouagadougou: about 1,200 km
- Ouagadougou–Niamey: about 500 km
- Niamey–Cotonou: about 900 km
- Cotonou–Lomé: about 150 km
- Lomé–Ouagadougou: about 850 km

The whole loop will not be finished until the 2030s.



Link: <https://pp2.au-pida.org/approved-project/entry/gvax6/>

Other West African lines include Guinea's CTG railway (the Mahina railway), linking the port of Maribaya with the Simandou mining area, about 552–600 km, officially opened in October 2025, designed for annual capacity of 220 million tonnes — the core hub for developing this world-class iron ore deposit. Built by a Chinese company.

On the whole, West African railway construction, except Nigeria, is progressing more slowly than East Africa's.

4. Road construction in West Africa

Road construction in West Africa is relatively fast, with the following key projects:

- The coastal expressway from Côte d'Ivoire's capital to Lagos, Nigeria — the key section of the West African transnational coastal highway, connecting five countries, about 1,000 km, six lanes both ways. By 2035, the population along this route will exceed 50 million. Construction is proceeding by sections; opening is expected before 2030.
- The Sierra Leone section of the transnational coastal highway, 480 km, two lanes both ways, mostly EU-funded with a small contribution from the African Development Bank.
- The Burkina Faso Ouagadougou–Bobo-Dioulasso expressway, 332 km, eight lanes both ways, connecting the country's two largest cities, launched by the recently installed nationalist "revolutionary" government, now under construction.
- Nigeria's Lagos–Calabar coastal highway, six to ten lanes both ways, crossing nine coastal states, 700 km, forming a transport artery together with the coastal railway. It will connect with the West African coastal highway at Lagos.
- Nigeria's Sokoto–Badagry Superhighway, six lanes both ways, 1,068 km, already under construction, planned for completion in 2031. When finished, road travel from Sokoto to Lagos will fall from 17 hours to about 8.
- Nigeria's Abuja–Kaduna–Zaria–Kano road, two lanes both ways, 375 km, nearly complete, expected within 2026.

Besides these, other roads of around 100 km are under construction or planned. By 2035, the road network of West Africa will be greatly improved.



The coastal expressway from Côte d'Ivoire's capital to Lagos, Nigeria

(2) Modernizing Energy Infrastructure

1. Electric power infrastructure

The figure below ^[31] shows per-capita electricity consumption trends in China, India, and sub-Saharan Africa. In the 1990s the gap among the three was small but existed, and sub-Saharan Africa was even higher than India. After the 1990s, China's per-capita consumption rose rapidly; after 2005, India's also rose fast, now reaching China's level of the early 2000s. Africa's per-capita consumption, however, remains very low.

Electric power consumption (kWh per capita) - Sub-Saharan Africa, China, India

IEA Energy Statistics Data Browser, International Energy Agency (IEA), uri: [iea.org/data-and-statistics/data-tools/energy-statistics-data-browser](https://www.iea.org/data-and-statistics/data-tools/energy-statistics-data-browser), publisher: International Energy Agency (IEA), date accessed: 2025-03-25

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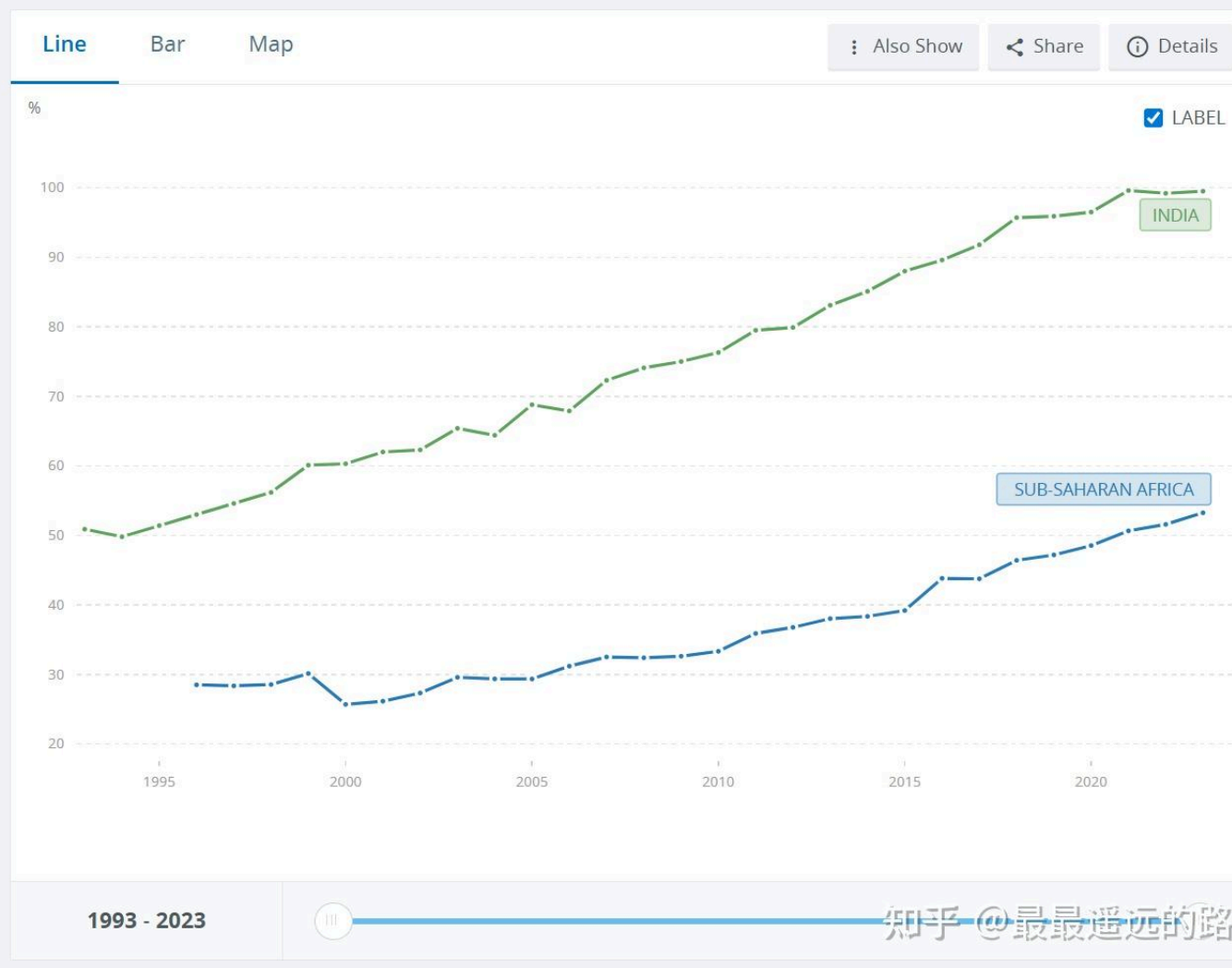
Power supply is the foundation of industrial development. Raw materials, components, and machinery can all be imported from industrial countries like China, Europe, and India — but electricity must be produced at home, or at least in the region. The power supply problem greatly constrains the speed of capital expansion in Africa. African countries are therefore working hard to rapidly increase power supply.

Since 2005, electricity access in sub-Saharan Africa has risen from 20 percent to over 50 percent, reaching India's level of the mid-1990s ^[32] — showing that African countries have made real efforts in power plant and grid construction. But because the African population grows very fast, per capita terms show little progress.

Access to electricity (% of population) - Sub-Saharan Africa, India

SDG 7.1.1 Electrification Dataset, World Bank (WB), uri: trackingsdg7.esmap.org/downloads, note: Data is downloaded from ESMAP website. Data is released when a new Tracking SDG7 report is released., publisher: World Bank (WB), date accessed: 2024-05-16, date published: 2023

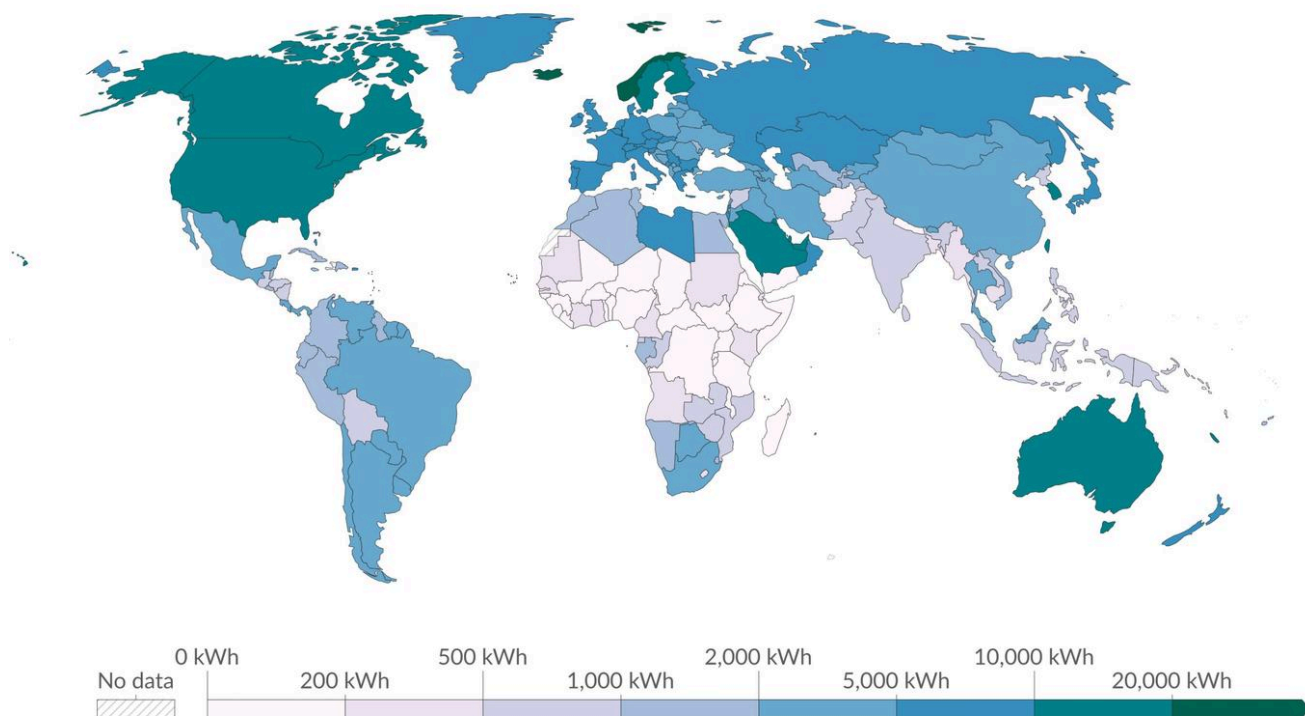
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In per-capita consumption [33] [34], East African countries have progressed rapidly in the last decade: Ethiopia rose from a few dozen kWh per year to over 200; Kenya from over 100 to over 200 — basically catching up with China's 1980 level. In West Africa, because population growth is very fast, per-capita consumption is stagnant except in Ghana, whose per-capita consumption has exceeded 600 kWh a year — China's level of the early 1990s — enough to support large-scale industrialization. Niger and several other backward West African countries use less than 100 kWh per capita per year; Nigeria is also at just over 100, with obvious supply problems. Among East African countries, Uganda and Tanzania also use less than 200 kWh per capita per year — clear power shortfalls. Southern Africa's generation is relatively high, mostly reaching China's 1990s-to-2000s levels.

Per capita electricity demand, 2015

Annual average electricity demand per person, measured in kilowatt-hours¹. This is electricity generation, adjusted for imports and exports.



Data source: Ember (2026); Population based on various sources (2024)

OurWorldinData.org/energy | CC BY

1. Watt-hour A watt-hour is the energy one watt of power delivers for one hour. Since one watt equals one joule per second, a watt-hour equals 3600 joules of energy.

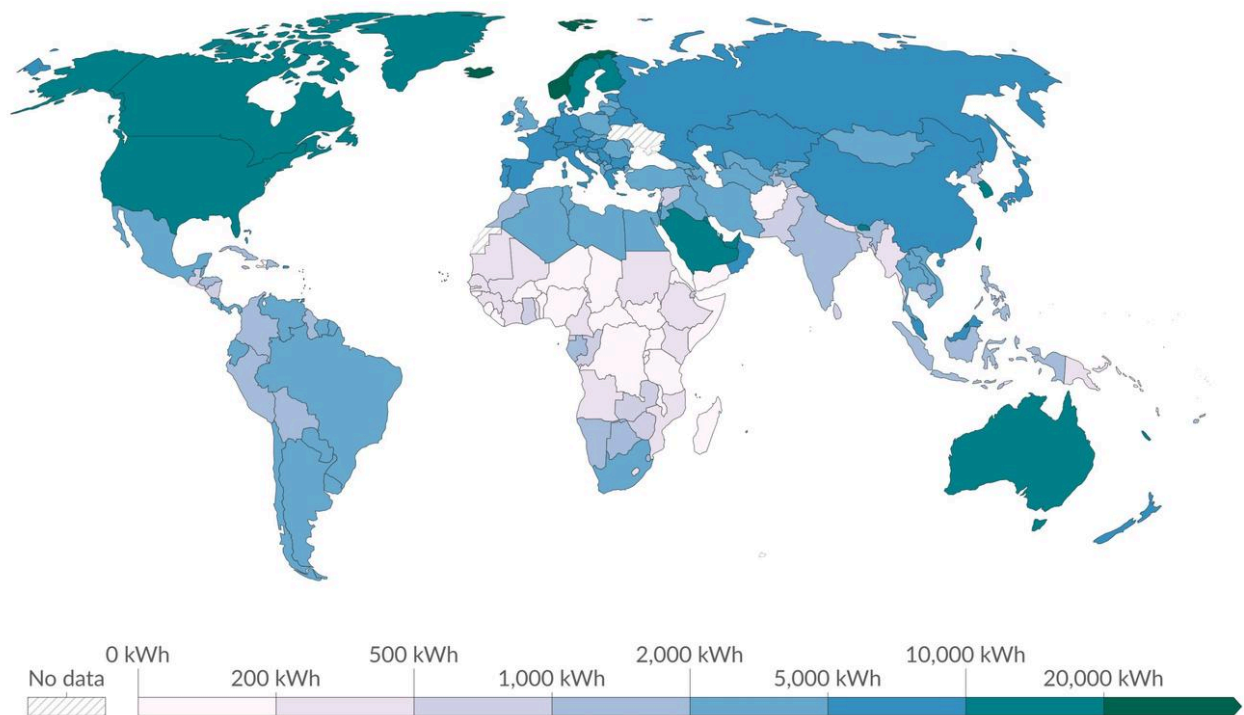
Metric prefixes are used for multiples of the unit, usually:

- kilowatt-hours (kWh), or a thousand watt-hours;
- Megawatt-hours (MWh), or a million watt-hours;
- Gigawatt-hours (GWh), or a billion watt-hours;
- Terawatt-hours (TWh), or a trillion watt-hours.

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Per capita electricity demand, 2025

Annual average electricity demand per person, measured in kilowatt-hours¹. This is electricity generation, adjusted for imports and exports.



Data source: Ember (2026); Population based on various sources (2024)

OurWorldinData.org/energy | CC BY

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Large-scale industrialization has begun in East Africa, and the power shortage is very evident; at present the region mainly relies on Ethiopia's Grand Ethiopian Renaissance Dam.

In addition, Kenya and Uganda are both preparing to build nuclear power plants. Kenya has signed with an American company to build a 2,000 MW nuclear plant ^[35] ^[36], with construction planned to start in 2027 and completion in 2034. This project will provide about 17.5 billion kWh a year — enough alone to raise Kenya's per-capita electricity use by about 250 kWh, bringing Kenya to China's 1990s level. Uganda also plans a 2 GW nuclear plant, most likely built by a Russian company. Beyond nuclear, East Africa is rich in geothermal resources, and the Kenyan government is vigorously developing them.

Tanzania is also building the large Julius Nyerere hydropower station, with total installed capacity of 2,115 MW and annual generation of 6.3 billion kWh, built by Dongfang Electric, which is also supplying all the equipment ^[37]. The station began commercial operation in June 2025. Tanzania is

also building some smaller solar and gas plants. The Tanzanian government recognizes that as industrialization advances, its power supply will be a major problem, and has begun a nuclear program — though it is still at an early stage, with construction probably in the 2030s.

Nigeria is prioritizing solar and gas power. It has already signed with Chinese firms for a 3 GW gas power station, a 4 GW wind-solar integrated plant, and hydropower projects, totaling 8 GW — 2.4 times the country's current power supply [38]. The Akwa Ibom gas combined-cycle plant has a total capacity of 3,000 MW, using efficient gas-steam combined-cycle technology, with 12 Siemens SGT5-2000E gas turbines, 12 waste-heat boilers supplied by Xizi Clean Energy, and 6 Siemens SST-600 steam turbines [39]. When complete, annual generation is expected at 15–20 billion kWh. Like the East African countries, Nigeria also has plans to develop nuclear power. Other West African countries, such as Ghana, have already signed nuclear contracts. At present, Russia, France, China, the US, and South Korea all have nuclear export capacity and compete fiercely, which works to African countries' advantage on cost and financing.

On the whole, within about ten years, per-capita annual electricity use in West and East Africa will basically exceed 500 kWh, with some countries possibly over 1,000 — reaching China's 1990s level, comparable to India's 2010 level [40].

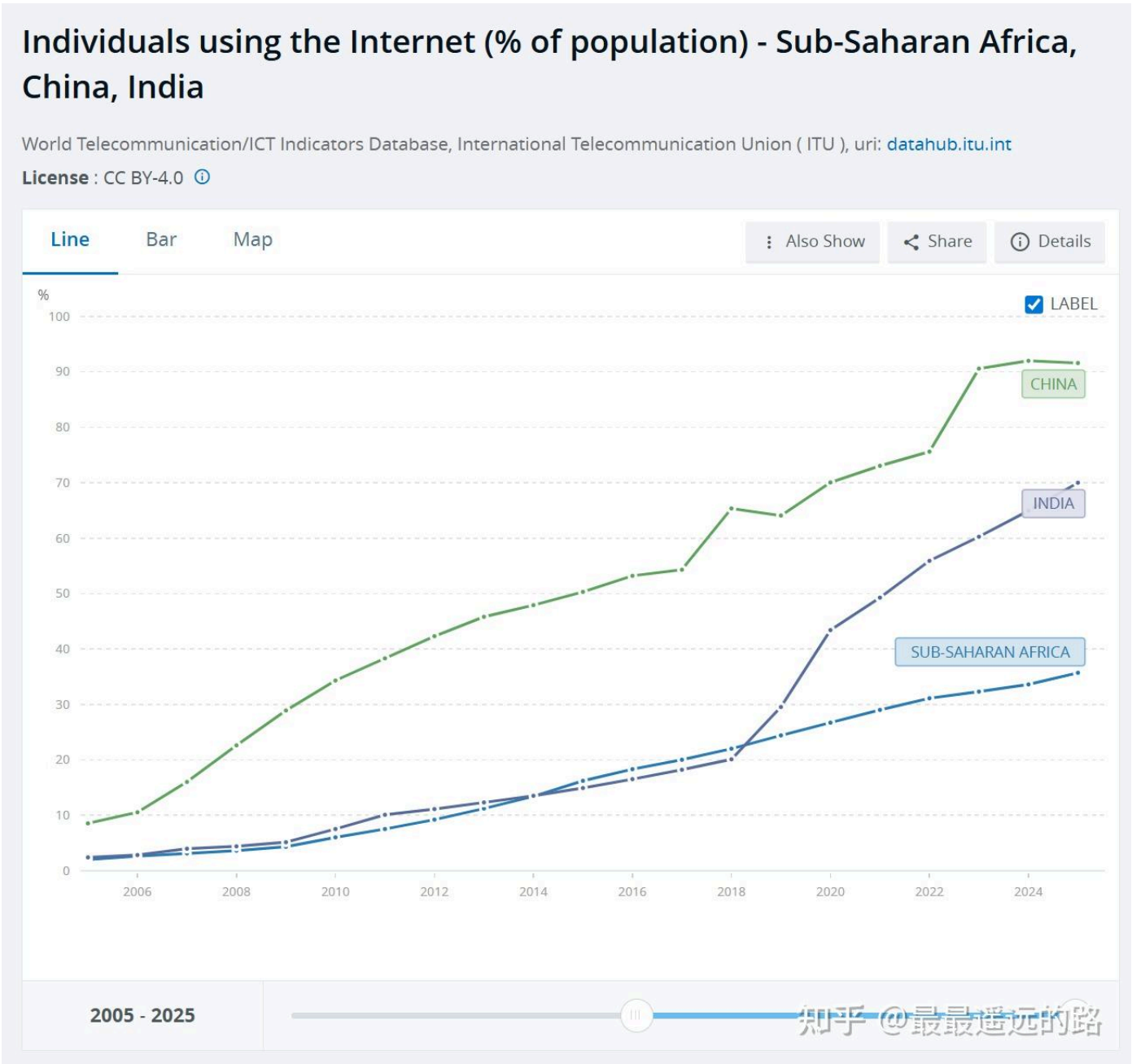
2. Oil refining infrastructure

Gasoline is an important energy source, roughly 30 percent of global primary energy consumption. Before the 2020s, although sub-Saharan Africa had many oil producers, its refining capacity was essentially negligible, requiring large gasoline imports that drained foreign exchange reserves.

In 2024, the Dangote Group's refinery came online at Nigeria's Lekki Free Trade Zone. Chinese companies did most of the construction and supplied the core equipment, while the design was a joint effort of Dangote Group and Indian firms. The project is designed to process 650,000 barrels of crude a day, with annual output of 850,000 tonnes of polypropylene and 500,000 tonnes of polyethylene resin, total investment over \$19 billion. As the world's largest single refinery, it ended Nigeria's long history of importing refined products and rewrote Africa's energy map. [41] Dangote now plans to expand the refinery to 1.4 million barrels a day, raising polypropylene output from 1 million to 1.5 million tonnes [42]. Dangote also plans a 650,000-barrel-a-day refinery in East Africa to serve the whole East African market; it is still deciding between Kenya and Tanzania [43]. When all these projects are complete, total crude processing will reach 2.05 million barrels a day — 100 million tonnes of crude a year, essentially China's 1990 level. Adding other refinery projects, by the early 2030s sub-Saharan refining capacity is expected to reach about 3.2 million barrels a day — about 158 million tonnes a year, close to India's 2010 level.

(3) Communications Infrastructure

The figure below shows the share of population using the internet in China, India, and sub-Saharan Africa [44]. Africa's internet usage lags India by only a few years. Network infrastructure requires far less investment than roads and rails, and with the spread of new technologies like Starlink [45], internet penetration in Africa will accelerate further.



The figure below [46] shows mobile subscriptions per 100 people in China, India, and Africa — sub-Saharan Africa even exceeds India. However, 5G is basically universal in the key areas of India's large and medium cities [47], while Africa's coverage is still very low [48].

Mobile cellular subscriptions (per 100 people) - Sub-Saharan Africa, India, China

World Telecommunication/ICT Indicators Database, International Telecommunication Union (ITU)

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According to some institutions' statistics ^[49], Africa's 4G coverage has already exceeded 75 percent — even discounted to 50 percent, that's not bad. In the last two years, sub-Saharan Africa has begun investing in 5G; Nigeria, Kenya, and other relatively developed countries have started using it. Ericsson estimates that over the next six years, sub-Saharan 5G subscribers will grow from 27 million to 400 million ^[50]. For Huawei, ZTE, and other telecom firms, this is a huge market.

Population coverage, by mobile network technology

Connectivity, Access

Refers to the percentage of inhabitants that are within range of at least a 4G/ LTE mobile-cellular signal, irrespective of whether or not they are subscribers

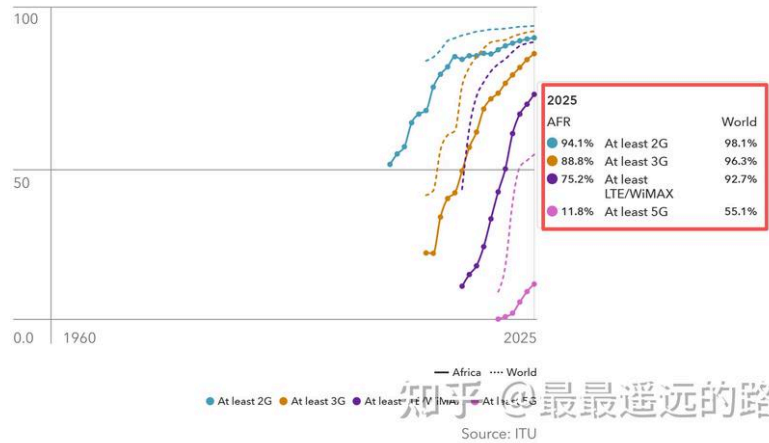
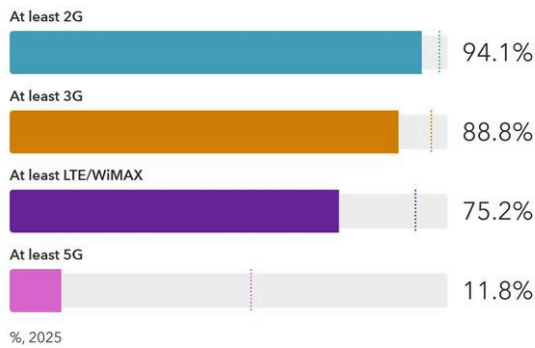
Overview

View: 

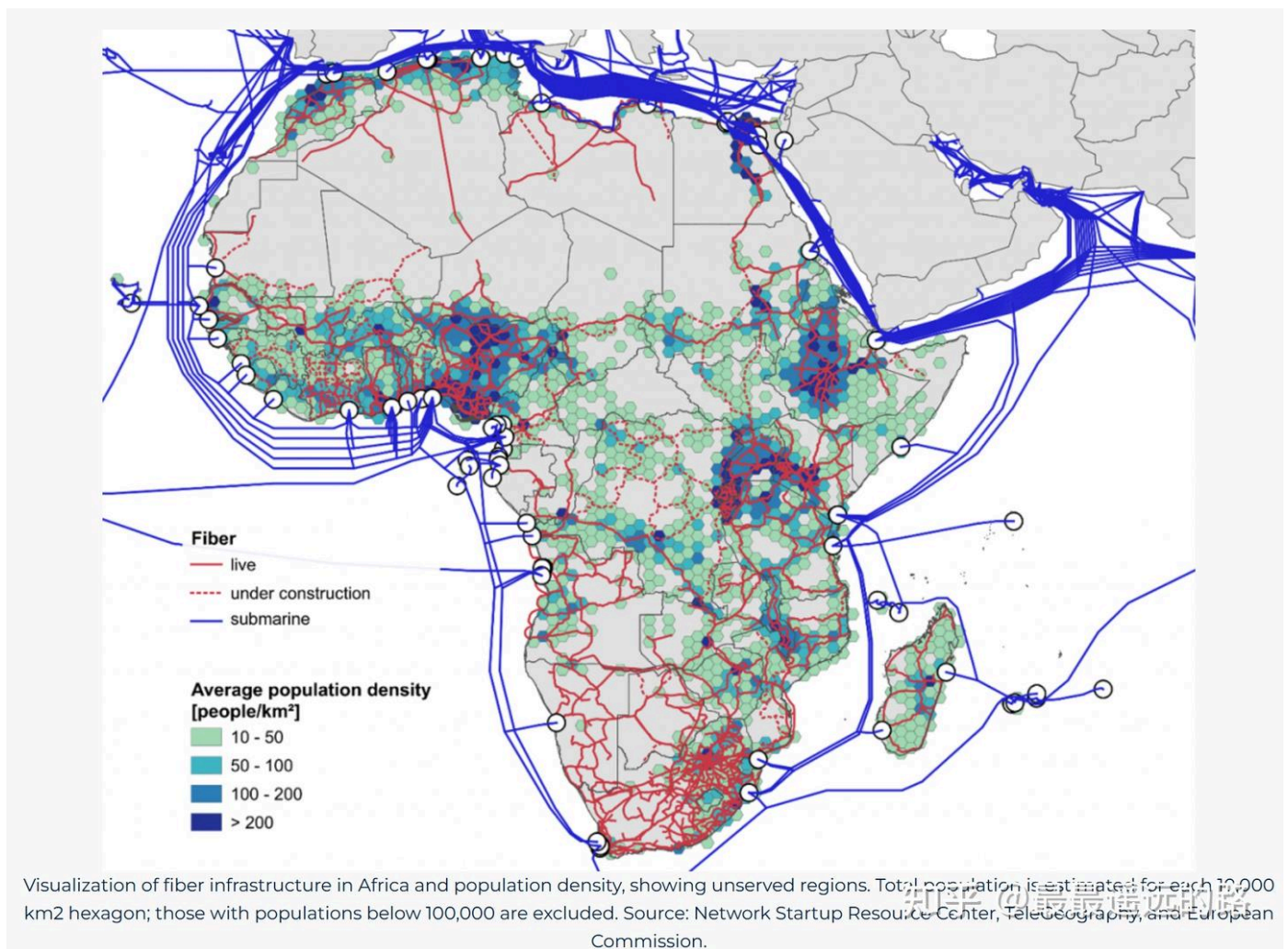


Unit: %

Download 



Africa's fiber-optic networks are also growing rapidly. Below [\[51\]](#) is a map of African fiber lines — red lines built, dashed lines under construction. Southern Africa, Nigeria, and East Africa (Kenya, Uganda) are basically connected; DR Congo and elsewhere are laying fiber on a large scale.



On the whole, Africa's communications infrastructure gap is being filled quickly; within ten years, except for the poorest countries like Niger, communications in the medium and large cities should essentially be solved.

III. Basic Industry in Sub-Saharan Africa

(1) The Cement Industry

Cement is the main material of construction; countries in rapid urbanization and industrialization have huge demand for it. In 1990, China's cement output was about 200 million tonnes. Sub-Saharan Africa's cement output is now approaching 100 million tonnes.

- Dangote Cement, a Nigerian firm, has plants in many African countries, with total output over 50 million tonnes.
- Lafarge Africa, mainly producing in Nigeria, output 10.5 million tonnes, expanding to 14 million.

- Bamburi Cement, Kenya's main cement maker, annual capacity about 1.8 million tonnes, expanding to 4 million.
- Devki Cement, another major Kenyan maker, annual capacity about 2.7 million tonnes, expanding to 7.5 million.
- Twiga Cement, a Tanzanian local firm, annual output 2 million tonnes. Tanzania's total annual cement capacity is about 11 million tonnes.
- West China Cement, a representative of Chinese enterprises' African footprint, owns plants in Zambia, Mozambique, Ethiopia, and other countries through acquisitions and new projects, with annual capacity of about 8 million tonnes.
- Ciments du Sahel, a Senegalese firm, annual capacity about 4 million tonnes.

Sub-Saharan Africa's large-scale urbanization and industrialization are driving rapid expansion of the cement industry; by around 2035, total sub-Saharan cement capacity can reach or even exceed China's 1990 level.

(2) The Steel Industry

The figure below shows steel output data of recent years. Only these few sub-Saharan countries are counted; China and India are shown for comparison. Kenya and Tanzania's steel output is rising fast, and Nigeria's has nearly doubled. Kenya and Nigeria have exceeded 1 million tonnes, and Tanzania is close to 1 million. Kenya's per-capita steel output already approaches China's 1990 level (about 50 kg).

Country	2021	2022	2023	2024	2025 (千吨)	2025年人口 (千)	2025相比 2021	2025年人均钢产 量(公斤)
Angola	275.0	270.0	190.0	285.0	250.0	39,000	0.91	6.41025641
China	1,035,242.6	1,017,959.0	1,028,859.7	1,005,091.3	960,810.0	1416000	0.93	678.5381356
Ghana	400.0	400.0	220.0	245.0	250.0	35,100	0.63	7.122507123
India	118,201.4	125,377.2	140,760.7	149,420.9	164,894.0	1463900	1.40	112.6402077
Kenya	485.0	450.0	1,365.0	1,750.0	1,800.0	57,500	3.71	31.30434783
Nigeria	650.0	650.0	1,190.0	1,200.0	1,200.0	238,000	1.85	5.042016807
South Africa	5,019.5	4,495.6	4,953.1	4,716.9	4,496.1	64,700	0.30	59.49149923
Tanzania	145.0	140.0	695.0	750.0	750.0	70,500	5.17	10.63610787

数据来源: <https://worldsteel.org/>

Other large steel projects under construction — besides Chinese firms, Indian and African local firms are also investing:

- Kenya's Devki Group (an Indian-origin company) has begun building a 1-million-tonne steel plant in Uganda, planned to start production in 2027.

- Anhui Tuoxin Steel's Maganga-Matitu iron ore project in Tanzania. The Tanzanian government holds 36 percent; China's Fujian Hexingwang Industrial (Tanzania) holds 64 percent. The Maganga-Matitu project has reserves of over 100 million tonnes of iron ore, planned to produce sponge iron with annual capacity of 1 million tonnes; full development was originally scheduled to begin in January this year, with production 18 months later [\[53\]](#) [\[52\]](#)
- On November 17, 2025, China Heavy Machinery signed an EPC contract for a 500,000-tonne-per-year integrated steel facility in Cameroon. The project will build a 500,000-tonne-per-year blast furnace steel plant in southern Cameroon, mainly producing bar and wire rod. As Cameroon's first long-process steel mill, it will fill the local steel capacity gap, provide stable steel supply for infrastructure construction, build a complete industrial chain, and drive coordinated development of related fields, injecting new momentum into local industry [\[54\]](#).
- Inner Galaxy Steel announced a \$550 million expansion, building a new 2-million-tonne steel plant in Nigeria's Ogun State [\[55\]](#).
- According to Zimbabwe News, Zimbabwe's steel exports have exploded, thanks mainly to Chinese enterprise empowerment and local capacity building. The steel plant's parent, China's Tsingshan Holding Group, announced an additional investment of about \$800 million (about 5.5 billion yuan) for supporting facilities and blast furnaces, raising carbon steel capacity from 600,000 to 1.2 million tonnes a year [\[56\]](#).
- On June 14, 2025, the Nigerian federal government announced a strategic cooperation with Chinese steel giants to restart the Ajaokuta Steel Company, stalled for decades. Steel Development Minister Shuaibu Audu led a delegation to China for investment promotion, holding technical assessment, operation modeling, and financing framework talks with senior executives of Sinosteel (a Baowu subsidiary), Fangda Steel, and Jingye Steel. Analysis notes the plant's design capacity of 1.9 million tonnes a year (expandable to 5 million) is regarded as a core asset for Nigeria's industrialization strategy [\[57\]](#).
- On April 15, 2026, Nigeria's Steel Development Minister Shuaibu Abubakar Audu and India's Laxmi Metal Group signed an MOU in Kolkata, with the Indian firm committing \$1 billion (about 6.8 billion yuan) of investment in Nigeria's steel industry over the next three years, helping Nigeria toward its 2030 goal of 10 million tonnes of crude steel capacity and making it the steel production center of West Africa [\[58\]](#).

Sub-Saharan Africa's steel industry is at the beginning of a great explosion; within 10 to 15 years, total sub-Saharan steel capacity is expected to reach China's 1990 level of about 50 million tonnes.

IV. Industries Integrated into the Global Value Chain

(1) Oil and Gas Production

Africa's oil and gas resources have also entered a phase of large-scale development — especially as the Iran war pushes more capital toward Africa.

Africa's upstream oil and gas industry is entering a stable phase; by 2026, daily output is expected to reach 11.4 million barrels of oil equivalent, with capital expenditure of about \$41 billion. Offshore deepwater oil and gas remains the main growth driver; as mature onshore basins face natural decline and higher reinvestment thresholds, offshore deepwater is increasingly key to long-term supply resilience. Justin Cochrane, S&P Global's Africa regional research director, stressed the generally low level of African exploration, noting that only about 25,000 wells have been drilled on the whole continent to date. He emphasized that since 2010, 74 percent of oil and gas discoveries have come from deepwater and ultra-deepwater areas, with gas accounting for 73 percent of total oil-and-gas discoveries. However, he cautioned that the commercialization of oil-and-gas resources remains uneven, with many frontier discoveries still lacking infrastructure or viable market access ^[59].

The main projects in Africa include:

Lake Albert project (Uganda): the Tilenga and Kingfisher projects are about 70–74 percent complete, with first oil expected in June 2026. Peak production is expected at about 190,000–230,000 barrels a day. Uganda has also built a pipeline from the oilfields to a Tanzanian port, designed to carry over 200,000 barrels of crude a day. As of May 2026, overall progress was 64 percent, with mechanical completion expected by the end of Q2 2026 and first oil in 2026. Baleine Phase 3 (Côte d'Ivoire): Eni's expansion aims to raise total output to 150,000 barrels a day by 2029. Bonga North (Nigeria): Shell's \$5 billion deepwater development, targeting first oil by the end of this decade. Nigeria's Bonga North deepwater project is designed for a peak of about 110,000 barrels of oil equivalent a day. Venus discovery (Equatorial Guinea): expected to reach about 15,000 barrels a day in 2026. The Nigeria–Morocco Gas Pipeline (NMGP), also called the Atlantic–Africa gas pipeline, is a massive project of about 5,600–6,900 km, designed to carry gas from Nigeria to North Africa and potentially to Europe. The Trans-Saharan Gas Pipeline (TSGP), also called the Niger gas pipeline, is a proposed 4,128-km, \$13 billion project to deliver up to 30 billion cubic meters of gas a year from Nigeria's delta region through Niger to Algeria's Mediterranean coast, ultimately for export to Europe. The project was restarted in February 2026, with construction expected to begin after Ramadan, aiming to connect with Algeria's existing Transmed pipeline to Italy.

Through large-scale oil-and-gas development, African countries can earn stable annual foreign exchange for importing the capital goods their industrialization requires.

(2) Mining

Besides oil and gas, African mining is also experiencing great development. Africa's mineral resources are extremely rich, with many minerals holding more than half of world reserves:

Platinum and palladium reserves hold 89 percent of the world; manganese 83 percent; chrome 91 percent. Platinum and palladium are concentrated in South Africa; chrome in South Africa and Zimbabwe. Gold, phosphates, cobalt: each over 50 percent of world reserves. The Atlas region is one of the world's three major phosphate producers; South Africa's gold output ranks first in the world. Gold mines are mainly in South Africa, Ghana, Zimbabwe, DR Congo, Tanzania, and Mali; phosphates are concentrated in Morocco, a famous phosphate producer; cobalt in DR Congo and Zambia. Uranium, tantalum, cesium, zirconium, graphite, bauxite: each over 30 percent of world reserves, of which Equatorial Guinea's bauxite holds 27 percent of the world. Uranium is concentrated in South Africa; tantalum in Mozambique; graphite in Madagascar; bauxite in Guinea, South Africa, Botswana, Guinea-Bissau, Ghana, Zimbabwe, etc. Iron, copper, zinc, bauxite: each over 20 percent of world reserves. The outer ring of the Congo Basin is the main metallic mineral zone; the copper mines of DR Congo and Zambia are famous for large reserves and high grade. West Africa mainly has iron ore and bauxite; Liberia is famous for iron ore ^[60].

With the entry of foreign capital, including Chinese, many world-class mining projects have been developed, such as Guinea's Simandou iron ore, and the copper and cobalt mines of Zambia and DR Congo.

The Simandou iron ore deposit, in southeastern Guinea, is the world's largest and highest-grade (iron content over 65 percent) undeveloped open-pit hematite deposit, with resources of over 4 billion tonnes. It is a strategic landmark project of China–Africa cooperation, jointly developed by Baowu Group, Chinalco, Rio Tinto, and the Winning Consortium. Simandou is one of the world's highest-quality, largest mining projects, with total investment of over \$20 billion, covering mines, railways, ports, and other systematic works. The delivered works include over 600 km of new multi-purpose trans-Guinea railway and supporting port facilities. The project formally entered production in November 2025; at full capacity, annual output can reach 120 million tonnes, of great significance in reducing China's dependence on Australian and Brazilian iron ore. ^[61] ^[62]

Zambia's Mingomba copper mine will cost over \$2.3 billion, one of the largest investments in Zambian history. When in production, it will produce over 300,000 tonnes of copper a year, ranking among Africa's top copper mines ^[63]. DR Congo's Sicomin copper-cobalt mine, with reserves of 8.68 million tonnes of contained copper metal, is a world-class super-large copper-cobalt mine; the project's total investment is \$6.7 billion, the largest non-energy Chinese investment in Africa ^[64].

African countries also realize that selling unprocessed minerals does not serve their industrialization, and have issued mining restrictions one after another, requiring foreign firms to process locally and raising their own value-added in the value chain. Media report that in the last year, more than a

dozen African countries have imposed mineral export bans to push foreign firms to build plants locally:

In February 2025, DR Congo banned raw cobalt exports, lifting the ban in October in favor of an annual export quota system. In June 2025, Gabon announced a total ban on raw manganese exports from January 1, 2029. In June 2025, Rwanda suspended new mineral trading licenses to review and improve industry regulations, covering tin, beryllium, lithium, etc. In October 2025, Malawi's presidential decree took effect immediately, banning all raw mineral exports, covering most minerals including lithium, rare earths, graphite, gold, and copper. In November 2025, Guinea raised bauxite export tariffs to 10 percent and mandated alumina refinery construction. In January 2026, Zambia passed "localization regulations" requiring mining companies to raise local procurement of core goods and services to at least 40 percent within specified years. In February 2026, Ghana's President Mahama announced plans to stop exporting any unprocessed ore — including manganese, bauxite, and iron ore — by 2030. In February 2026, Zimbabwe brought its ban forward, suspending all raw and concentrate lithium exports from February 25, with in-transit cargo also controlled. In March 2026, Senegal revoked 71 mining licenses on a large scale, covering various domestic minerals. In April 2026, Kenyan President Ruto announced a total ban on exports of all primary minerals — gold, iron, copper, rare earths, etc. — stepping up local refining. In April 2026, Mali's new Mining Code took effect, requiring the state to automatically receive a 10 percent free carried interest and to be able to buy up to an additional 20 percent at par within two years of commercial production. In May 2026, Mozambique's parliament passed amendments to the Mining Law, banning raw mineral exports and requiring state shareholding of no less than 15 percent. ^[65]

In the future there will only be more such policies. African countries have awakened; they realize the minerals in their hands are precious resources that must be used well. Through promoting localized production, mandatory state shares or equity participation, they can raise export value, increase government revenue, and provide more funds for national industrialization.

(3) Light Industry

Besides oil, gas, and minerals, another African industry integrating into the global value chain is light industry. Using Africa's cheap land and labor to export agricultural processed goods or light industrial products — garments, leather — to the industrialized countries in exchange for foreign exchange is an important part of sub-Saharan Africa's current industrialization stage. Below we mainly discuss agricultural products and garments; other industries serving mainly the domestic market are not discussed here.

1. Agricultural product exports

Moving from simply exporting agricultural products to processing them is the main way African countries climb the value chain:

Cocoa beans are the core raw material of chocolate. Côte d'Ivoire is the world's largest cocoa producer, and neighbor Ghana second, with only a third of Côte d'Ivoire's output. Nearly two-thirds of global cocoa comes from these two West African countries. Cocoa exports remain an important foreign exchange source for Côte d'Ivoire: nearly half of its beans are exported as raw material, the other half semi-processed. The Ivorian government keeps encouraging localization of cocoa processing and supports investment in plants. At the PK24 industrial park northwest of Abidjan, a cocoa processing plant and bean warehouse contracted by China Light Industry Nanning Design and Engineering Co. are nearing completion. When finished, it will become Côte d'Ivoire's largest domestically owned cocoa processing plant, converting beans into finished products like cocoa butter and cocoa powder. The Ivorian government has high hopes for the industry, aiming for 100 percent local processing of cocoa beans by 2030 ^[66]. Coffee is one of Kenya's key economic pillars, long among the top earners of agricultural export revenue. As China's coffee market keeps expanding, Kenya has become one of China's important coffee import sources. In a corner of the plant, Mbulah deliberately showed the reporter Chinese coffee roasting equipment from the factory. In recent years, while continuing green-bean exports, Utake has also been improving local processing capacity. "If we can export roasted coffee, we can create more jobs, let more trained people participate, and add value," she said ^[67]. Rwanda's dried chilies are more than four times hotter than ordinary dried chilies. Since entering the Chinese market in 2021, they have become a fine, cost-effective ingredient for hotpot base and braising spices, loved by domestic consumers. From site selection, seedling, planting to harvesting, processing, and export, the cooperation project covers every link of the chili supply chain. The 100-hectare demonstration farm exported 200 tonnes of dried chilies to China in batches in its first harvest season. The cooperation has also changed local farmers' lives: the farm now has 15 employees, hiring over a thousand temporary workers in the chili harvest season ^[68].

With China's zero tariffs on nearly all African countries, exporting agricultural and agro-processed products to China will become an important way for Africa to earn foreign exchange. Apart from machinery and equipment, agricultural processing needs no imported raw materials; with Africa's low wages and China's zero tariffs, opening farms and processing plants in Africa is a worthwhile investment for capitalists.

2. The garment industry

East Africa's garment industry already has a certain scale. Chinese enterprises' African factories mainly export to Europe and America.

"There are plenty of opportunities in Africa; at the peak, I didn't even know which one to take." Gu Jingyong, chairman of C&D Apparel from China, says that since settling in Rwanda four years ago, the company has grown from one factory to four, with nearly 5,000 workers, becoming Rwanda's largest garment factory and Africa's largest jacket factory. "Orders are too many; we work every day, even Sundays." C&D's four African factories produce 2.5–3 million garments a year, 90 percent exported to

Europe and America, with export sales of about 50 million yuan. Gu stressed this scale would be unremarkable at home, but in Africa it makes them a benchmark and leader. As overseas capacity climbs, Shengtai Group is enjoying an explosion of globalization dividends. In recent years, its international footprint has moved from Asia to Africa. At the Eastern Industry Zone near Ethiopia's capital it has a 10-million-piece-a-year knitted garment plant, and at the Huajian International Light Industry Park near the capital, a 10-million-piece-a-year woven garment plant ^[69].

In 2012, as domestic labor costs rose and the international trade environment grew complicated, Changzhou Dong'ao Garments built a factory in Tanzania's Benjamin William Mkapa Export Processing Zone, about 15 km from the economic capital Dar es Salaam. It now has seven garment factories, providing over 7,000 local jobs, and has become Tanzania's largest garment processing enterprise. Its products are mainly denim, exported duty-free to the US market under AGOA; customers include LEVIS, WRANGLER, and PRIMARK. Electricity costs 3 cents per kWh; workers' base wages are \$80 a month ^[70].

There have been cases of Chinese garment factories in Southeast Asia exporting back to China. As China's labor force further shrinks and Southeast Asian labor costs climb higher, more firms will move to East Africa. At that point, China will import garments from Africa in large quantities.

V. Education Levels in Sub-Saharan Africa

The figure below ^[71] shows primary school enrollment in China, India, and sub-Saharan Africa. Sub-Saharan Africa experienced a great expansion of primary education after 2000, and now basically matches India's level of the 2000s.

School enrollment, primary (% gross) - India, Sub-Saharan Africa, China

Data API, UN Educational, Scientific and Cultural Organization (UNESCO), uri: databrowser.uis.unesco.org/resources, note: The data are obtained through the UIS API. Detailed documentation is available at: api.uis.unesco.org/api/public/documentation, publisher: UNESCO Institute for Statistics (UIS), type: Bulk file (csv), date accessed: 2026-03-19, date published: 2026-02

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The figure below ^[72] shows secondary enrollment in China, India, and sub-Saharan Africa. Here too, sub-Saharan Africa has reached India's 2000s level. In 2010, about 35 million children were born in sub-Saharan Africa, so Africa can now add at least 15 million workers a year with basic literacy. As primary and secondary education spread further and births increase (45 million in 2025), this number will rise to about 30 million.

School enrollment, secondary (% gross) - China, India, Sub-Saharan Africa

Data API, UN Educational, Scientific and Cultural Organization (UNESCO), uri: databrowser.uis.unesco.org/resources, note: The data are obtained through the UIS API. Detailed documentation is available at: api.uis.unesco.org/api/public/documentation, publisher: UNESCO Institute for Statistics (UIS), type: Bulk file (csv), date accessed: 2026-03-19, date published: 2026-02

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The figure below ^[73] shows tertiary enrollment in China, India, and sub-Saharan Africa. Higher education generally spreads only after large-scale industrialization begins. Africa's higher-education expansion is expected after 2035, similar to India's after 2010. India's tertiary enrollment is expected to exceed 30 percent by around 2045. In 2025, sub-Saharan Africa had about 45 million births, so around 2045 Africa can add 13.5 million university graduates a year — comparable to China's current level. By the end of 2050, when enrollment rises further to about 50 percent, Africa can add over 20 million university graduates a year.

School enrollment, tertiary (% gross) - Sub-Saharan Africa, China, India

Data API, UN Educational, Scientific and Cultural Organization (UNESCO), uri: databrowser.uis.unesco.org/resources, note: The data are obtained through the UIS API. Detailed documentation is available at: api.uis.unesco.org/api/public/documentation, publisher: UNESCO Institute for Statistics (UIS), type: Bulk file (csv), date accessed: 2026-03-19, date published: 2026-02

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VI. An Estimate of Sub-Saharan Africa's Industrialization Progress

As described above, Africa's industrialization is roughly 20 years behind India's. And as I analyzed in a series of earlier answers and articles, India's industrialization is itself roughly 20 years behind China's (see: "Is India seriously underrated domestically?"). However, the conditions facing African industrialization are far better than India's. First, because Africa is the last low-cost frontier of global labor, the competitive pressure on African industrialization is far smaller than India's. Second, because China and India have completed industrialization one after another, the suppliers able to transfer technology and capital goods to Africa have greatly multiplied. Third, because China is in the

final stage of transferring its low-end industries, the export of Chinese industrial capacity to Africa is accelerating; China's zero-tariff policy toward Africa serves precisely this goal.

(1) 2026–2035: The Last Ten Years of Laying the Foundation

Over the next decade, Africa will gradually complete its main railways, industrial lines, and oil-and-gas pipelines; oil refining and power generation will gradually reach China's 1990 or India's 2010 levels. Literacy and education will improve further. By 2035, sub-Saharan Africa will possess the basic conditions for receiving industrial transfer on a large scale. At the same time, with the AfCFTA gradually taking effect, Africa will become a super-large single market of nearly 2 billion people (rising to 2.5 billion by 2050) — enormously attractive to transnational capital.

(2) 2036–2045: Receiving Low-End Industry on a Large Scale

By 2035, all of China's 1960s-born workers will have left the labor market, half of the 1970s-born will have left, over 250 million non-university-educated workers will have exited, and about 200 million more will have withdrawn from high-intensity manual labor (under 50). This enormous labor demand cannot be met even by draining nearly all of Southeast Asia's surplus labor. Moreover, since Southeast Asian countries are also nearing the completion of their industrialization, labor costs are rising rapidly. So, whether by cost or by human constraints, Chinese firms will have to transfer to Africa. At the same time, as Southeast Asian and Indian industrialization deepens, the big enterprises of these regions will accelerate outward transfer, and Africa is almost the only destination for low-end industrial relocation. Africa adds over 30 million workers a year with at least secondary education; besides emigrating to Europe, America, and other countries to solve the labor crises of collapsing fertility in the destination countries, they will be absorbed by the industries transferred from China, India, Southeast Asia, and the Middle East. Notably, from the 2040s on, the major industrial regions — North America, Europe, and the whole of East Asia — will face annual labor shortfalls of nearly 15 million people or more. With fertility in almost all regions fallen below replacement level (consider: Turkey's fertility has fallen to 1.47, India's to 1.9, and Egypt's will fall below the replacement rate of 2.1 in 2030), a labor demand of this scale can probably only be supplied by sub-Saharan Africa.

(3) 2045–2055: The Difficult Rise of Mid- and High-End Manufacturing

As African education develops, the cohorts born in the 2020s and after will graduate from university and join various local technology and manufacturing enterprises, and Africa will gradually move up the value chain. However, as I analyzed in "Can the World Make Room for a Developed China?":

By 2050, the developed countries, China included, have fallen to 25.5 percent of the world — but the next wave of aspirant countries adds up to 24.8 percent. If India and others rise to developed status as

they hope, the non-agricultural working-age population of the developed countries would reach 50 percent of the world — which, by the historical benchmark of 2010, is plainly impossible. Count in the newly industrializing countries with real strength and ambition — Egypt, Bangladesh, Indonesia, Morocco, South Africa — and the share rises toward 60 percent, which is even more impossible. [74]

By the 2050s, even India — already upgrading its industries and facing fierce industrial competition from China, Europe, America, Japan, and Korea — will find it nearly impossible to complete the upgrade as earlier countries did; Africa, occupying mainly the middle and lower reaches of the production chain, faces even greater competitive pressure. But Africans are people too, and want a good life; African enterprises will inevitably push toward the upper reaches of the industrial chain in every sector. Transnational capital has two responses: first, gradually shift R&D and high-end manufacturing to Africa, using African human resources to cut costs and adapt more effectively to the African market; second, as African mid- and low-end manufacturing costs rise, gradually return some manufacturing to the mature industrial countries. The rise of India and Africa must therefore break the original structure of the world-system and push toward the equalization of the living standards of the proletariat of all countries. Besides facing the same multinational employers, the convergence of the living standards of the world's working classes is the key material condition for the genuine unification of the world proletariat. Thus, those of us still alive may well experience an extraordinary historical process in the late 2050s.

Notes

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管理水平的物流与IT设备。联合国粮农组织（FAO）数据显示，2025年埃塞俄比亚机械化耕作农田面积约500万公顷，占全国1840万公顷总耕地面积的27%左右。根据长期规划，埃塞政府计划将农业机械保有量扩大三倍以上：拖拉机数量预计从2025年初的2万台增至6.5万台，联合收割机从2700台增至1.5万台。（编选：中非贸易研究中心）

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